

# Assessing Misreading of Axis Scales and Units in AI-Based Chart Interpretation

## Abstract

As Multimodal Large Language Models (MLLMs) and Vision-Language Models (VLMs) become increasingly integrated into automated data analysis, their ability to accurately interpret data visualizations is paramount. Despite achieving high accuracy on standardized benchmarks, these models frequently struggle with the precise spatial grounding and numerical extraction required to read complex charts. This paper investigates a critical point of failure in AI-based chart interpretation: the systematic misreading of axis scales and units. By examining current literature on multimodal chart comprehension, this study highlights how features such as logarithmic scales, dual axes, non-standard units, and dense tick marks trigger explanation-level hallucinations and data misalignment. We review current interventions—such as plot-to-table translation modules and sketch-based reasoning frameworks—and outline the persistent research gaps in evaluating explanation faithfulness over mere answer correctness.

**Keywords:** Multimodal Large Language Models (MLLMs), Chart Understanding, Vision-Language Models (VLMs), Hallucination, Axis Misinterpretation, Data Visualization

## 1. Introduction

Charts and plots are high-density visualization carriers for complex data, serving as crucial mediums for information extraction in scientific, financial, and business contexts. The advent of Multimodal Large Language Models (MLLMs), such as GPT-4o, Gemini, and open-weight equivalents like InternVL and Qwen-VL, has catalyzed a shift toward automated chart interpretation. These systems can answer queries, summarize trends, and extract data directly from bitmap images of charts.

However, extracting semantic meaning from charts necessitates more than natural language processing; it requires precise visual-spatial reasoning. A persistent vulnerability in state-of-the-art MLLMs is their inability to accurately read and map axis scales, tick marks, and units to the corresponding data series. Even when models arrive at the correct final answer, they frequently fabricate the intermediate steps of reading the axes—a phenomenon known as explanation-level hallucination. This paper provides a scholarly assessment of how and why AI models misread axis scales and units, evaluates current technical mitigations, and identifies critical limitations in how chart understanding is currently benchmarked.

## 2. Background and Literature Review

The evaluation of chart understanding in AI initially relied on template-based, procedurally generated datasets with limited visual diversity. Early benchmarks measured basic perceptual tasks, such as identifying chart types or retrieving explicit values.

The introduction of **ChartQA** by Masry et al. (2022) marked a paradigm shift by incorporating human-authored, complex reasoning questions that require both visual and logical synthesis. ChartQA revealed that while models could identify general trends, they struggled significantly with tasks requiring arithmetic operations based on visual estimations from axes.

Recent literature underscores that standard evaluation suites often overestimate model capabilities. Zhang et al. (2024) introduced **CharXiv**, a benchmark derived from real-world scientific papers, revealing a substantial, previously underestimated gap between proprietary models (e.g., GPT-4o achieving 47.1% accuracy) and human performance (80.5%) on reasoning tasks. A primary driver of this performance gap is the failure of models to reliably interpret structural elements like legends, units, and axes. Furthermore, Chen et al. (2024) demonstrated in their assessment of misleading visualizations that LLMs are vulnerable to misinterpreting spatial representations when axes are manipulated.

### **3. Main Thematic and Technical Discussion**

The misreading of axes by MLLMs is not a monolithic error but manifests through several specific visual-reasoning failures.

#### **3.1. The Vulnerability of Logarithmic Scales**

Standard linear axes are generally well-comprehended by top-tier MLLMs because the spatial distance between tick marks correlates directly and proportionally with numerical values. However, logarithmic scales heavily disrupt the heuristic spatial assumptions of VLMs. When interpreting log-scaled axes, MLLMs frequently map data points using linear interpolation, leading to massive numerical extraction errors. This stems from the models' over-reliance on learned geometric patterns rather than reading the explicit mathematical relationship defined by the tick mark labels.

#### **3.2. Dual Axis Interpretability and Data Misalignment**

Dual-axis charts (charts containing a primary y-axis on the left and a secondary y-axis on the right) present a severe binding problem for VLMs. Models must not only read the axes but associate specific data series (e.g., a bar chart and a line plot superimposed) with the correct axis. Studies testing the vulnerability of LLMs to chart manipulation indicate that dual-axis configurations frequently lead to misinterpretation, as the model incorrectly maps the magnitude of one data series to the scale of the opposing axis. This failure in visual-semantic grounding results in models falsely claiming correlations or magnitude differences that do not exist in the underlying data table.

#### **3.3. Unit Conversion and Tick-Mark Hallucinations**

A frequent limitation in AI chart interpretation is the precise reading of dense numerical tick marks. When data points lie between explicit tick marks, humans use proportional spatial estimation. VLMs, however, often struggle to interpolate these values accurately. Lim et al. (2025) noted in their work on the ChartCap framework that "axis hallucinations" account for a significant portion of errors in dense chart captioning.

Furthermore, units of measurement (e.g., millions, percentages, specific scientific metrics) are often presented in abbreviated forms on the axis label. If the model fails to optically extract or semantically prioritize the axis title, it defaults to raw numerical outputs, resulting in a scale magnitude error (e.g., reporting "5" instead of "5 million").

### **4. Current Approaches and Developments**

To circumvent the visual-spatial limitations of purely end-to-end MLLMs, the field has seen several innovative developments aimed at bridging the gap between pixel space and numerical

reasoning.

#### 4.1. Modality Conversion: Plot-to-Table Translation

One of the most effective interventions is bypassing direct visual reasoning through modality conversion. Liu et al. (2023) introduced **DePlot**, a one-shot visual language reasoning module that translates the image of a plot into a linearized data table. By first derendering the chart into structural text, the system relies on the robust numerical and logical reasoning capabilities of standard LLMs to answer queries, effectively bypassing the visual axis-reading bottleneck.

#### 4.2. Visual Grounding and Sketch-Based Reasoning

Other approaches attempt to fix the visual reasoning process itself. **ChartLlama** (Han et al., 2023) utilized specialized instruction-tuning datasets to improve the model's fundamental grounding of chart elements. More recently, Li et al. (2025) proposed **ChartSketcher**, which addresses the limitation of textual Chain-of-Thought (CoT) by enabling MLLMs to explicitly annotate intermediate reasoning steps directly onto the chart image. By drawing on the image (e.g., projecting a line from a data point to the y-axis to read the scale), the model leverages visual feedback and reflection, mimicking the human subconscious strategy of tracing lines to axes for accurate value estimation.

#### 4.3. Multi-Agent and Verification Frameworks

To address temporal and trend hallucinations, frameworks like **ChartInsighter** (Wu et al., 2025) utilize multi-agent iterative collaboration and self-consistency testing. By combining visualization specifications with the underlying data tables, these systems cross-verify the LLM's generated summary against the raw data, explicitly mitigating extremum errors and trend direction hallucinations.

### 5. Research Challenges and Limitations

Despite advancements, evaluating and mitigating axis and scale misinterpretations faces profound structural challenges within the research community.

#### 5.1. Explanation-Level Hallucination

As VLMs increasingly produce free-form explanations for chart reasoning, researchers have discovered a critical flaw: models can provide the correct final answer while entirely fabricating the axis-reading steps. Kim et al. (2026) defined this as "explanation-level hallucination". In a study of ChartQA instances, they found that 87.1% of QA-correct responses from base VLMs contained claims unsupported by visual evidence—meaning the model guessed or inferred the correct answer without faithfully reading the axis or tick marks.

#### 5.2. Benchmark Blindness

This high rate of hallucination exposes a limitation known as "Benchmark Blindness". Standard evaluation suites grade models primarily on answer extraction accuracy (often using a relaxed numerical threshold). Because the evaluation function projects only the final token and discards the reasoning, benchmark scores are inflated, masking the model's fundamental inability to read scales and units reliably.

### 6. Research Gaps and Future Directions

The current trajectory of chart interpretation research suggests several critical gaps that must

be addressed:

1. **Faithfulness as an Independent Metric:** Future benchmarks must adopt claim-level verification to ensure that intermediate visual reasoning (e.g., reading an axis intercept) is as faithful as the final answer.
2. **Specialized Datasets for Non-Linear Scales:** Existing training corpora are heavily biased toward simple, linear bar and line charts. There is a pressing need for curated datasets focusing explicitly on dual axes, logarithmic scales, and broken axes to enforce robust spatial grounding.
3. **Integration of Metadata and Source Verification:** As seen in studies of biological and scientific charts, LLMs often hallucinate conclusions based on their training priors rather than the visual evidence. Future systems must better separate the extraction of visual facts (what the axis actually says) from the model's semantic priors (what the model expects the axis to say).

## 7. Conclusion

The automated interpretation of charts by Multimodal Large Language Models holds transformative potential for data analysis. However, as this paper demonstrates, the precise extraction of numerical data via axis scales and units remains a fragile point in the AI reasoning pipeline. MLLMs consistently struggle with logarithmic scales, dual-axis misalignment, and tick-mark interpolation, often hiding these perceptual failures behind explanation-level hallucinations. While innovations like plot-to-table translation (DePlot) and multimodal feedback (ChartSketcher) show great promise in mitigating these errors, the community must evolve its evaluation metrics. Until benchmark blindness is resolved and explanation faithfulness is explicitly optimized, the vulnerability of AI models to misreading axis scales will remain a significant barrier to their autonomous deployment in critical analytical settings.

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